

Telangana Single File DB1 Pipeline Documentation

This Python script implements a comprehensive data processing pipeline for Telangana DB1 (Database 1) property transaction data. The pipeline processes individual Excel files through multiple stages including data cleaning, property type classification, RERA project matching, BHK assignment, geocoding, and final data standardization.

The pipeline transforms raw property transaction data from Telangana into a structured, analysis-ready format by:

- Cleaning and normalizing property descriptions
- Extracting key information (dates, parties, document details)
- Classifying property types using regex patterns
- Matching projects with RERA master data
- Assigning BHK configurations
- Geocoding locations (optional)
- Creating standardized output files
 - pandas
 - numpy
 - re (regex)
 - ast
 - datetime
 - pathlib
 - argparse
 - rapidfuzz (for fuzzy matching)
 - tqdm (for progress bars)
 - geopy (for geocoding, optional)
 - **Input Excel File:** Single Telangana DB1 Excel file to process
 - **RERA Master File:** Telangana RERA grand Excel file for project matching
 - --input-file: Path to input Excel file (required)
 - --rera-file: Path to RERA master Excel file (required)
 - --output-dir: Output directory (default: "telangana_single_file_output")
 - --enable-geocoding: Enable ArcGIS geocoding for locations (optional, slow)
 - **■ Load Input File:** Reads the specified Excel file into memory
 - **■ Validate Files:** Checks existence of input and RERA master files
 - **■ Create Output Directory:** Sets up folder structure for results
 - **■ Display Configuration:** Shows processing parameters and settings

- **■ Duplicate Removal:** Eliminates duplicate records across key columns
- **■ Invalid Record Removal:** Removes rows containing only "-" or "W-B: 0-0"
- **■ Index Reset:** Resets DataFrame index after cleaning operations

Extracts structured information from property descriptions:

- **■■ Village/Colony (VILL/COL):** Location identifiers
- **■■ Ward-Block (W-B):** Administrative divisions
- **■ Survey Number:** Land survey identifiers
- **■ Plot Number:** Individual plot identifiers
- **■ House Number:** Building identifiers
- **■ Apartment/Project Name:** Complex/project names
- **■ Block:** Building block identifiers
- **■ Flat/Unit Number:** Individual unit identifiers
- **■ Extent (area):** Land area measurements
- **■■ Built-up area:** Constructed area measurements
- **■■ Boundaries:** Property boundary descriptions
 - **■ Dates:** Registration, Execution, and Presentation dates
 - **■ Document Information:** Document type code, type, market value, consideration value
 - **■ Parties:** Seller and Buyer names with party type classification
 - **■ Transaction Type:** Sales, Lease, or Others categorization

Uses regex patterns to classify properties into:

- **■ Flat/Apartment:** Residential units in complexes
- **■ Shop:** Commercial retail spaces
- **■ Office:** Commercial office spaces
- **■ House:** Independent residential buildings
- **■ Parking:** Parking spaces and areas
- **■ Plot/Land:** Vacant land parcels
- **■ Others:** Unclassified property types
 - **■ Extent Conversion:** Converts SQ.Yd to SQ.Ft (multiplies by 9)
 - **■■ Built-up Area Formatting:** Standardizes built-up area values
 - **■ Final Area Calculation:** Determines area based on property type logic

Computes price per square foot using the formula:

Performs fuzzy matching between processed data and RERA master file:

- **■ Combined Match:** Project name + location matching
- **■■ Project-Only Match:** Project name only matching

- ■ **Assigns RERA Data:** Index, project details, coordinates, and pincode

For flat properties, assigns BHK configuration by:

- ■ **Area Matching:** Matches final area with RERA BHK-wise carpet areas
- ■ **Percentile Ranges:** Uses statistical ranges for fallback assignment
- ■ **Match Differences:** Calculates and validates area differences

Uses ArcGIS geocoder to obtain latitude/longitude for locations:

- ■ **Unique Processing:** Processes each location only once
- ■ **Result Caching:** Stores results to avoid duplicate API calls
- ■ **Coordinate Updates:** Adds lat/long to location data
 - ■ **Title Case Conversion:** Standardizes text formatting
 - ■ **Year/Quarter Extraction:** Creates time period columns from dates
 - ■■ **Column Renaming:** Applies standard column naming conventions
 - ■ **Checklist Creation:** Generates data quality validation report

The pipeline generates multiple output files in the specified output directory:

- **deleted_records.xlsx:** Records removed during cleaning
- **cleaned_dataset.xlsx:** Dataset after initial cleaning
- **extraction_output.xlsx:** Data after property type classification
- **test.xlsx:** Intermediate test output
- **Telangana_Final_With_Rate.xlsx:** Data with rate calculations
- **matched_output.xlsx:** Data after RERA matching
- **output_clean.xlsx:** Cleaned output with selected columns
- **output_with_bhk_Assign.xlsx:** Data with BHK assignments
- **geocoded_location_output.csv:** Geocoded location data (if enabled)
- **Updated_title_case_output.xlsx:** Title case formatted data
- **Year_Quarter_output.xlsx:** Data with year/quarter columns
- **Telangana_DB1_With_Year_Quarter_output_Updated.xlsx:** Final standardized output
- **telangana_checklistDB1.xlsx:** Data quality checklist
- **segment_fields():** Parses property description text
- **extract_dates():** Extracts date information
- **extract_document_info():** Parses document details
- **extract_parties():** Identifies sellers and buyers
- **classify_property_type_regex():** Classifies property types
- **normalize_light():** Normalizes text for matching
- **get_best_combined_match():** Fuzzy matches project + location
- **get_best_project_only_match():** Fuzzy matches project name only
- **compute_location_similarity():** Validates location matches

- `extract_bhk_from_key()`: Parses BHK from RERA data
- `assign_bhk_from_final_area()`: Assigns BHK based on area matching
- `assign_bhk_fallback()`: Uses percentile ranges for fallback

The pipeline includes comprehensive validation:

- Column presence verification
- Data type checking
- Null value analysis
- Unique value counts
- Sample value inspection
- Mapping verification
 - **Geocoding**: Can be slow due to API rate limits; processes unique locations only
 - **Fuzzy Matching**: Uses efficient algorithms but may be time-consuming for large datasets
 - **Memory Usage**: Processes data in memory; monitor for large files
 - Validates input file existence
 - Handles missing columns gracefully
 - Provides fallback logic for missing data
 - Continues processing despite individual record errors
 - Strong Match: $\geq 90\%$ similarity
 - Good Match: $\geq 85\%$ similarity
 - Review: $\geq 80\%$ similarity
 - Maximum area difference: 5 square meters
 - Uses percentile-based ranges for fallback
 - **Missing Dependencies**: Install required packages
 - **File Not Found**: Verify input file paths
 - **Memory Errors**: Process smaller files or increase system memory
 - **Geocoding Failures**: Check internet connection and API limits
 - Check input file format and columns
 - Verify RERA master file structure
 - Review output logs for errors
 - Validate final output against expectations
 - Update regex patterns as new property description formats emerge
 - Refresh RERA master data periodically
 - Monitor geocoding API usage and costs
 - Review matching thresholds based on accuracy needs
 - Initial implementation based on `Telangana_DB1_Pipeline.ipynb`
 - Added comprehensive error handling and logging

- Implemented fuzzy matching for improved accuracy
- Added BHK assignment and geocoding features
- Created standardized output format with checklist

Note: This documentation is generated from the code analysis. For detailed implementation specifics, refer to the source code comments and function docstrings.